

ME381 ROBOTICS

Experiment No.: 3

Experiment Title: Sensor interfacing with microcontroller

Group No.: B6

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Roll no.: 230793

Results for the group task:

Task no.	Topic	Task	Report
1	Button (digital)	Make the connections as shown in the 'Task Manual' and upload the program for Task 1. Press the button and observe the state of built-in LED. Modify the code to change the logic in such a way that LED turns 'off' when button is pressed. It should remain 'on' whenever button is in 'released' position.	(a) When button is not pressed, LED is ON, and when it is pressed, it turns OFF. (b) In the if statement, I changed the argument of digitalWrite from "HIGH" to "LOW", and inside else statement, I changed the argument of digitalWrite from "LOW" to "HIGH"
2	Hall sensor (digital)	Make the connections as shown in the 'Task Manual' and upload the program for Task 2. Move a magnet across the hall sensor module in swiping fashion and observe the LED on the module and the Serial Monitor. Turn the magnet such that the other pole now faces the hall sensor. Repeat the same swiping motion and observe the LED on the module and the Serial Monitor.	(a) The default output of hall sensor in the absence of any magnet is HIGH. (b) When one side of the magnet was facing the sensor, the LED in the sensor is ON and output on the serial monitor is LOW, and when other side faces, sensor LED turns OFF and output in the serial monitor is HIGH.
3	Infrared (IR) transceiver pair (digital)	Make the connections as shown in the 'Task Manual' and upload the program for Task 3. Bring any obstacle (preferably light in color with smooth texture) in front of the IR transceiver. Adjust the potentiometer on the module to adjust the range of IR transceiver.	(a) When an obstacle is there in front of the IR transceiver, the LED state on the IR module is ON and the LED state on the Arduino is OFF. (b) The default output of the IR transceiver when it is not sensing any obstacle is "HIGH".
4		Make the connections as shown in the 'Task Manual' and upload the program for Task 4.	(a) The pulse count shown in the Serial Monitor after 5 full

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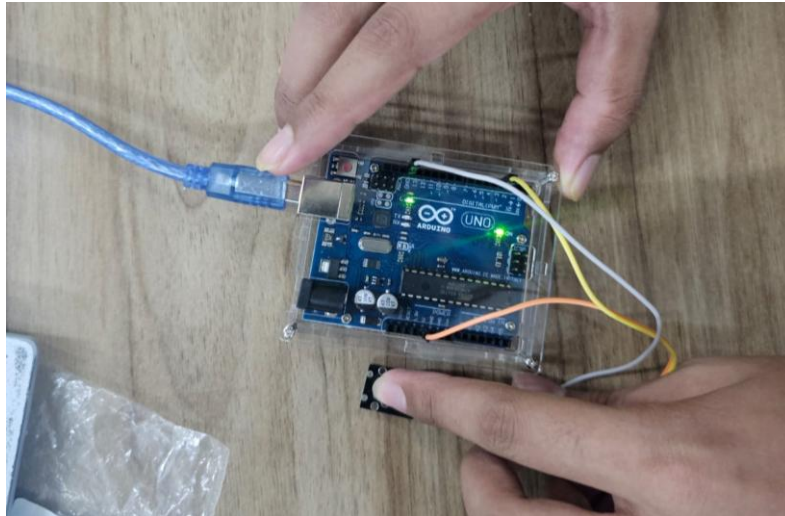
	Inductive proximity sensor (digital)	RESET the microcontroller once. Now rotate the rotor using your hands slowly. Make sure not to reverse the direction of rotation in between.	rotations of the rotor is 15.
5	Potentiometer (analog)	Make the connections as shown in the 'Task Manual' and upload the program for Task 5. Observe the Serial monitor while rotating the knob of the potentiometer.	(a) When the potentiometer knob is turned to its end limit in clockwise direction then "Sensor Value: 0" and "Mapped Value: 0". (b) When the potentiometer knob is turned to its end limit in counterclockwise direction then "Sensor Value: 1023" and "Mapped Value: 255".
6	Incremental encoder (digital)	Make the connections as shown in the 'Task Manual' and upload the program for Task 6. Rotate the knob of the incremental encoder in one direction and observe the readings on the Serial Monitor. Reverse the direction of rotation of the knob and observe the readings on the Serial Monitor.	(a) There are no limits on the rotation of the knob and it can be rotated infinitely in any of the directions. (b) When the direction of rotation is clockwise, encoderValue increases, and when direction is reversed, it decreases. (c) Starting from zero, when the knob is given one full rotation, "60" gets displayed on the Serial Monitor. Potentiometer X value: 512 Mapped brightness: 127 (d)
7	Joystick with RGB LED (analog)	Make the connections as shown in the 'Task Manual' and upload the program for Task 7. Observe the RGB LED module and the serial monitor while tilting the knob of the joystick along x and y directions.	(a) when the joystick knob is tilted to extreme -ve x direction, the potentiometer X value is 506 and the colour of the LED is "Blue". (b) when the joystick knob is tilted to extreme +ve y direction, the potentiometer Y value is 526 and the colour of the LED is "Red".

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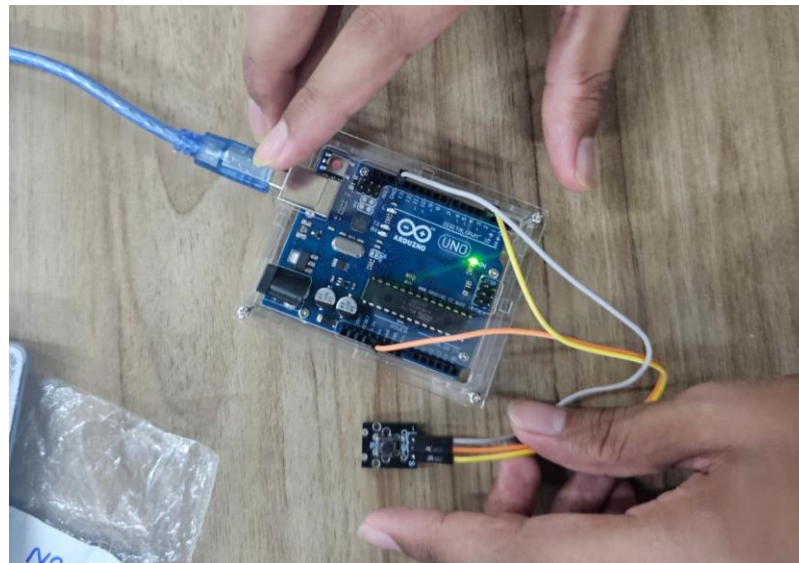
Results for individually assigned task:

Task: 1

(a).

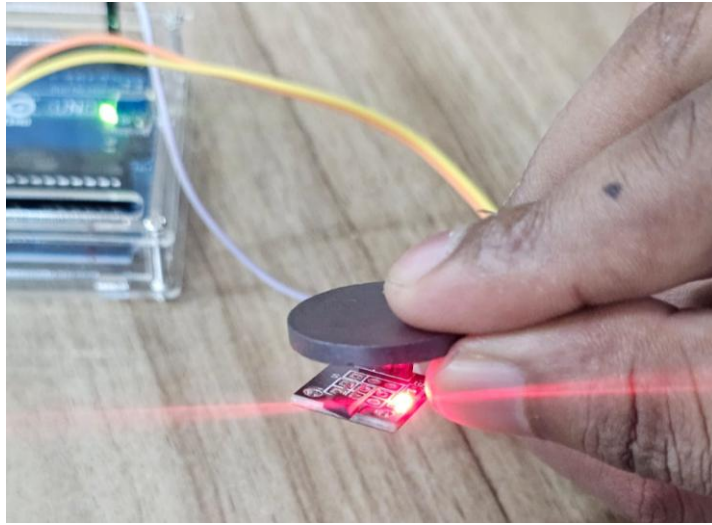


(b).

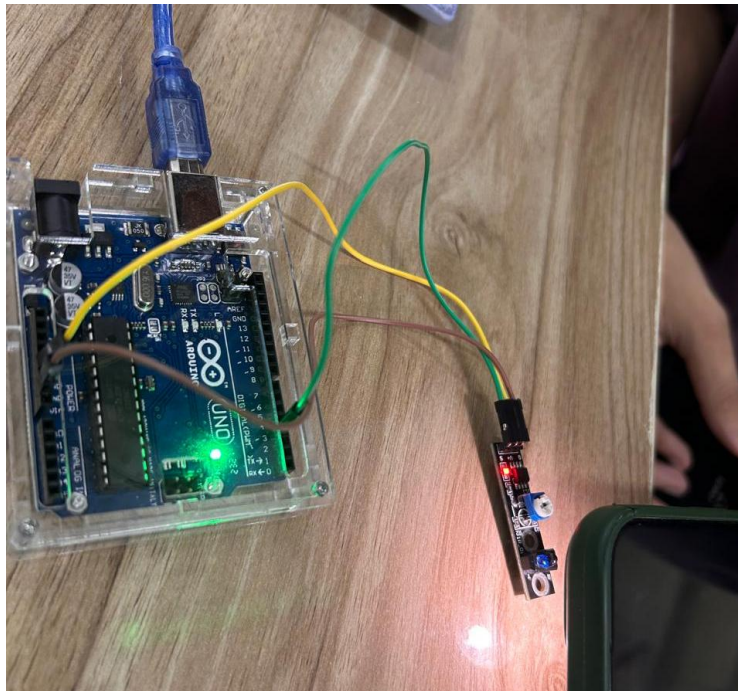


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Task: 2



Task: 3



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Task: 4

[illegible]

Task: 5

(a).

```
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0  
Sensor Value: 0  
Mapped Value: 0
```



(b).

[illegible]

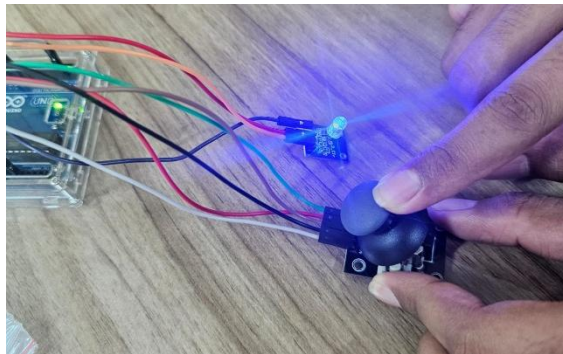
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Task:6

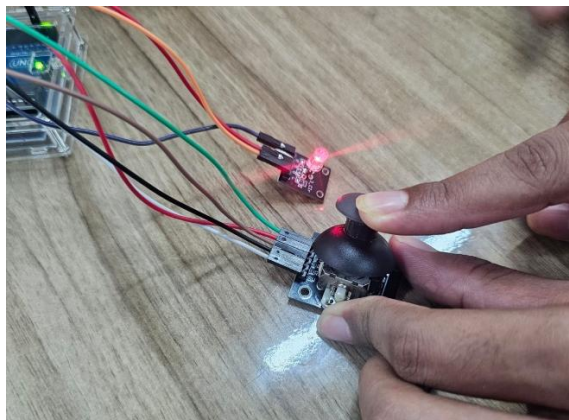


Task:7

(a).



(b).



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